

Project Report On

Online Movie Ticket Booking System

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF COMPUTER APPLICATION

With

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by:

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Under the Guidance of

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CERTIFICATE

This is to certify that the project report entitled “**Online Movie Ticket Booking System**” submitted by *[Your Name]* (University Roll No. _____) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Application (BCA)** at **Graphic Era Hill University, Bhimtal** is a record of bonafide work carried out by him/her under my supervision and guidance.

To the best of my knowledge, this project work has not been submitted elsewhere for the award of any other degree or diploma.

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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the Project Report entitled “**Online Movie Ticket Booking System**” in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Computer Application (BCA)** of **Graphic Era Hill University, Bhimtal** has been carried out by the undersigned under the supervision of **Dr. Janmejy Pant, Assistant Professor, School of Computing, Graphic Era Hill University, Bhimtal.**

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The above mentioned student(s) has successfully completed the project work under my supervision entitled “**Online Movie Ticket Booking System.**”

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Place: Bhimtal

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ABSTRACT

In today's fast-paced digital era, the demand for online services has significantly increased, especially in the entertainment industry. Traditional methods of booking movie tickets involve long queues, time consumption, and inconvenience for users. To overcome these challenges, an efficient and user-friendly system is required that allows users to book movie tickets quickly and conveniently from anywhere.

The **Online Movie Ticket Booking System** is designed to provide a digital platform where users can view available movies, check show timings, select seats, and book tickets in an efficient manner. The system simplifies the ticket booking process by eliminating the need for physical presence at cinema halls and ensures a smooth and hassle-free experience for users.

The proposed system is developed using **Python programming language** with **Tkinter** for graphical user interface design, while **MySQL** is used for database management. The system stores important user data, booking details, and seat availability information, ensuring accurate and real-time updates. It provides an interactive interface where users can easily navigate through different options and complete the booking process within a few steps.

The application includes features such as movie selection, seat availability display, ticket confirmation, and database storage of booking records. The system ensures efficient data handling, reduces manual errors, and improves overall user experience. Additionally, it demonstrates the integration of frontend and backend technologies along with database connectivity.

This project highlights the practical implementation of software development concepts in solving real-world problems. It aims to provide a reliable, fast, and user-friendly solution for movie ticket booking. In the future, the system can be enhanced by integrating online payment gateways, mobile application support, and real-time updates from cinema servers to further improve functionality and user satisfaction.

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1. Introduction

1.1 Introduction

In today's digital world, the use of online systems has become an essential part of everyday life. With the rapid growth of the internet and technology, people prefer online platforms for various services such as shopping, banking, and entertainment. One such important area is movie ticket booking. Traditional ticket booking methods involve standing in long queues at cinema halls, which is time-consuming and inconvenient for users.

The **Online Movie Ticket Booking System** is designed to overcome these limitations by providing a digital solution that allows users to book movie tickets easily from anywhere. The system provides a user-friendly interface where users can view available movies, select show timings, choose seats, and confirm bookings in a simple and efficient manner.

1.2 Background of the Study

Earlier, movie tickets were booked manually at cinema counters, which required physical presence and often resulted in long waiting times. With the advancement of web technologies and software systems, online booking platforms have been introduced to improve user convenience. Popular systems such as online ticket booking websites have transformed the way people access entertainment services.

The integration of programming languages like Python and database systems like MySQL has made it possible to develop efficient booking systems. These systems provide real-time information about movie availability, seating arrangements, and booking status. This project is based on the concept of automating the ticket booking process using software technology to improve efficiency and accessibility.

1.3 Problem Statement

The traditional method of booking movie tickets manually creates several problems such as long queues, time wastage, limited availability of tickets, and lack of real-time information. Users often face inconvenience due to delays and uncertainty in seat availability.

There is a need for an automated system that can provide real-time booking services, reduce manual effort, and ensure a smooth and efficient ticket booking experience.

1.4 Objectives of the Project

The main objectives of the Online Movie Ticket Booking System are:

- To develop a user-friendly application for booking movie tickets
 - To provide real-time seat availability and booking status
 - To reduce manual effort and waiting time
 - To store booking details securely in a database
 - To ensure accuracy and efficiency in the booking process
 - To design an interactive and easy-to-use interface
-

1.5 Scope of the Project

The scope of this project is to provide a simple and efficient system for booking movie tickets. The system allows users to select movies, choose seats, and confirm bookings through a graphical interface. It can be used by cinema halls to manage ticket bookings and by users to reserve seats conveniently.

The project mainly focuses on basic booking functionalities and demonstrates the use of software technologies in solving real-world problems. In future, the system can be expanded by adding features such as online payment, mobile application support, and real-time updates from cinema servers.

1.6 Organization of the Report

This project report is organized into various chapters to provide a clear understanding of the system. The report begins with an introduction to the project, followed by literature review, system analysis, and system design. Further chapters explain the methodology, implementation, testing, and results of the system. The report concludes with features, requirements, conclusion, future scope, and references.

2. Literature Survey

2.1 Introduction

Literature review is an important part of any project as it provides an understanding of existing systems, technologies, and methodologies related to the proposed system. It helps in identifying the limitations of current solutions and provides a base for developing an improved system.

In the context of movie ticket booking, various traditional and modern systems have been used to facilitate the process. This chapter discusses the existing approaches, technologies used, and their limitations, which justify the need for an efficient Online Movie Ticket Booking System.

2.2 Existing Systems

Earlier, movie ticket booking was completely manual, where customers had to visit cinema halls and stand in long queues to purchase tickets. This process was time-consuming and inconvenient, especially during weekends and new movie releases.

With the advancement of technology, online booking platforms were introduced. These systems allow users to browse movies, check show timings, and book tickets online. Popular applications and websites provide these services with additional features such as seat selection, payment gateways, and booking confirmation.

Most existing systems are web-based and use technologies like HTML, CSS, JavaScript for frontend development, and backend technologies such as PHP, Python, or Java. Databases like MySQL are used to store user information and booking records. These systems provide real-time updates and improve user experience significantly compared to manual systems.

2.3 Technologies Used in Existing Systems

Modern movie booking systems use a combination of technologies to ensure smooth functionality. Some of the commonly used technologies include:

- **Frontend Technologies:** HTML, CSS, JavaScript for designing user interfaces
- **Backend Technologies:** Python, Java, PHP for processing user requests
- **Database Systems:** MySQL, Oracle for storing booking data
- **GUI Frameworks:** Tkinter (for desktop-based applications)
- **Web Frameworks:** Flask, Django for web-based systems

These technologies help in building efficient, interactive, and scalable systems.

2.4 Limitations of Existing Systems

Despite improvements, existing systems still face certain limitations:

- Complex user interfaces that may confuse users
- Dependency on stable internet connection
- Security concerns in online payment systems
- Limited customization in some applications
- High system complexity in large-scale platforms

These limitations highlight the need for a simpler and more user-friendly system that focuses on essential booking functionalities.

2.5 Proposed System

The proposed **Online Movie Ticket Booking System** aims to overcome the limitations of existing systems by providing a simple, efficient, and user-friendly solution. The system is designed using Python and Tkinter for easy interaction and MySQL for secure data storage.

It allows users to:

- View available movies
- Select seats
- Book tickets easily
- Store booking details in database

The system focuses on simplicity, efficiency, and ease of use, making it suitable for academic and small-scale implementations.

2.6 Summary

From the literature review, it is clear that online booking systems have significantly improved the ticket booking process. However, there is still scope for developing systems that are simpler, efficient, and user-friendly. The proposed system builds upon existing technologies to provide a reliable and effective solution for movie ticket booking.

3. System Analysis

3.1 Introduction

System analysis is a crucial phase in the development of any software system. It involves understanding the requirements of the system, identifying user needs, and analyzing how the proposed system will function. This phase helps in defining the overall structure, features, and limitations of the system before actual implementation.

The **Online Movie Ticket Booking System** is analyzed to determine its functionality, performance, and feasibility. The aim is to develop a system that provides a smooth and efficient ticket booking experience to users while overcoming the limitations of traditional methods.

3.2 Requirement Analysis

Requirement analysis involves identifying the needs of users and defining the functionalities that the system must provide. It ensures that the developed system meets user expectations effectively.

3.2.1 User Requirements

The system should fulfill the following user requirements:

- Users should be able to view available movies and show timings
- Users should be able to select seats according to availability
- The system should allow ticket booking in an easy and quick manner
- Users should receive confirmation of their booking
- The system should provide a simple and user-friendly interface

3.2.2 System Requirements

The system must include the following technical requirements:

- A graphical user interface for interaction
 - Backend processing using Python
 - Database management using MySQL
 - Seat availability tracking mechanism
 - Data storage for booking records
-

3.3 Functional Requirements

Functional requirements describe the specific operations that the system performs. These include:

- Displaying list of movies and show timings
 - Allowing users to select seats
 - Validating user inputs
 - Processing booking requests
 - Generating booking confirmation
 - Storing booking details in database
 - Retrieving stored data when required
-

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system. These include:

- **Performance:** The system should process bookings quickly
 - **Usability:** The interface should be simple and easy to use
 - **Reliability:** The system should work without errors
 - **Security:** User data should be stored securely
 - **Scalability:** System should support future enhancements
 - **Compatibility:** Should work on different systems and platforms
-

3.5 Feasibility Study

Feasibility study determines whether the proposed system is practical and beneficial.

3.5.1 Technical Feasibility

The system uses technologies like Python, Tkinter, and MySQL, which are easily available and widely used. Development tools are simple and do not require advanced infrastructure.

3.5.2 Economic Feasibility

The project is cost-effective as it uses open-source technologies. No expensive software or hardware is required.

3.5.3 Operational Feasibility

The system is easy to operate and does not require technical expertise. Users can interact with the system easily through a graphical interface.

3.6 System Constraints

Some limitations of the system include:

- Requires basic computer knowledge
 - Limited to local system (if not web-based)
 - No online payment integration (in basic version)
-

3.7 Summary

System analysis helps in understanding the requirements and feasibility of the proposed system. The Online Movie Ticket Booking System is found to be technically, economically, and operationally feasible. It provides an efficient solution for ticket booking and meets all necessary user requirements.

4. System Design

4.1 Introduction

System design is the process of defining the architecture, components, modules, data flow, and database structure of the proposed system. It acts as a blueprint for the development phase and helps in converting system requirements into a working application.

The **Online Movie Ticket Booking System** is designed to provide a simple, efficient, and user-friendly platform for booking movie tickets. The system includes a graphical user interface, backend processing, and database management.

4.2 System Architecture

The system follows a simple **3-layer architecture**:

1. Presentation Layer

This layer is used for user interaction. It includes the graphical interface where users can view movies, select seats, and book tickets.

2. Application Layer

This layer handles the main processing of the system. It validates user input, processes booking requests, and manages seat availability.

3. Database Layer

This layer stores booking details, movie records, seat information, and user data using MySQL database.

4.3 Data Flow Diagram (DFD)

DFD Level 0

The overall flow of the system is:

User → Online Movie Ticket Booking System → Booking Confirmation

The user enters booking details, the system processes the request, checks seat availability, stores data, and displays confirmation.

DFD Level 1

Detailed flow:

1. User selects movie
 2. System displays show timings
 3. User selects seat
 4. System checks seat availability
 5. Booking details are stored in database
 6. Confirmation message is displayed
-

4.4 Flowchart

The flow of the system is:

Start

↓

Display Movie List

↓

Select Movie

↓

Select Show Time

↓

Choose Seat

↓

Check Seat Availability

↓

If Seat Available → Confirm Booking

↓

Store Booking in Database

↓

Display Ticket Confirmation

↓

Stop

4.5 Database Design

4.5.1 ER Diagram Explanation

The database contains important entities such as:

- **User**
- **Movie**
- **Seat**
- **Booking**

The user selects a movie and seat, and the booking details are stored in the database.

4.5.2 Database Table Structure

Table: movie_booking

Field Name	Data Type
id	INT PRIMARY KEY
user_name	VARCHAR(50)
movie_name	VARCHAR(100)
show_time	VARCHAR(50)
seat_no	VARCHAR(20)
ticket_price	INT
booking_status	VARCHAR(30)

4.6 UML Diagrams

The following UML diagrams can be used in the system design:

Use Case Diagram

Shows interaction between the user and the system.

Main actions are movie selection, seat selection, ticket booking, and confirmation.

Activity Diagram

Shows the step-by-step working of the booking process from movie selection to final confirmation.

Sequence Diagram

Shows communication between user interface, backend logic, and database during ticket booking.

4.7 Summary

The system design of the **Online Movie Ticket Booking System** provides a clear structure for implementation. It defines the architecture, data flow, database design, and system workflow. This design ensures smooth functioning, easy maintenance, and future enhancement of the system.

5. Methodology and Algorithm

5.1 Introduction

Methodology explains the step-by-step process followed to develop the **Online Movie Ticket Booking System**. It describes how user input is collected, processed, stored, and converted into a final ticket booking confirmation.

5.2 Development Methodology

The project follows a structured development approach:

1. Requirement Analysis
2. System Design
3. Database Design
4. GUI Development
5. Backend Implementation
6. Testing and Validation

This approach helps in developing the system in an organized and systematic manner.

5.3 Data Collection

The system collects the following details from the user:

- User Name
- Movie Name
- Show Time
- Seat Number
- Ticket Price
- Booking Status

These details are used to generate and store booking records.

5.4 Data Processing

After collecting data, the system checks whether the selected seat is available or already booked. If the seat is available, the booking is confirmed. If the seat is already booked, the system displays an error message and asks the user to choose another seat.

5.5 Working Methodology

The working process of the system is:

1. User opens the application
 2. Movie list is displayed
 3. User selects movie and show time
 4. Seat layout is displayed
 5. User selects available seat
 6. System checks seat availability
 7. Booking details are stored in database
 8. Confirmation message is displayed
-

5.6 Algorithm

1. Start
 2. Display movie list
 3. Accept user name
 4. Accept movie selection
 5. Accept show time
 6. Display available seats
 7. User selects seat
 8. Check seat availability
 9. If seat is already booked, display “Seat Not Available”
 10. If seat is available, confirm booking
 11. Store booking details in database
 12. Display ticket confirmation
 13. Stop
-

5.7 Summary

The methodology and algorithm explain the complete working of the **Online Movie Ticket Booking System**. The system follows a simple and efficient process to book tickets, check seat availability, store records, and display confirmation to the user.

6. Implementation

6.1 Introduction

Implementation is the phase where the designed system is converted into a working application. In the **Online Movie Ticket Booking System**, different modules such as user interface, movie selection, seat selection, booking confirmation, and database storage are developed and integrated together.

6.2 Frontend Implementation

The frontend is developed using **Python Tkinter**. It provides a simple graphical interface where users can enter their details and book movie tickets.

Frontend Features

- User name input field
 - Movie selection option
 - Show timing selection
 - Seat selection buttons
 - Booking confirmation message
-

6.3 Backend Implementation

The backend is developed using **Python**. It handles the main logic of the system.

Backend Functions

- Accept user input
 - Validate booking details
 - Check seat availability
 - Process ticket booking
 - Display confirmation message
-

6.4 Database Implementation

The system uses **MySQL** database to store booking records.

Database Operations

- Store user name
- Store movie name
- Store show timing
- Store seat number
- Store ticket price
- Store booking status

Example table used:

Field Name	Description
id	Unique booking ID
user_name	Name of user
movie_name	Selected movie
show_time	Selected show timing
seat_no	Selected seat
ticket_price	Price of ticket
status	Booking status

6.5 Module Implementation

1. Movie Selection Module

This module displays the available movies to the user. The user can select any movie from the list according to preference.

2. Show Timing Module

This module allows the user to choose suitable show timing for the selected movie.

3. Seat Selection Module

This module displays seat numbers and allows users to select available seats.

4. Booking Module

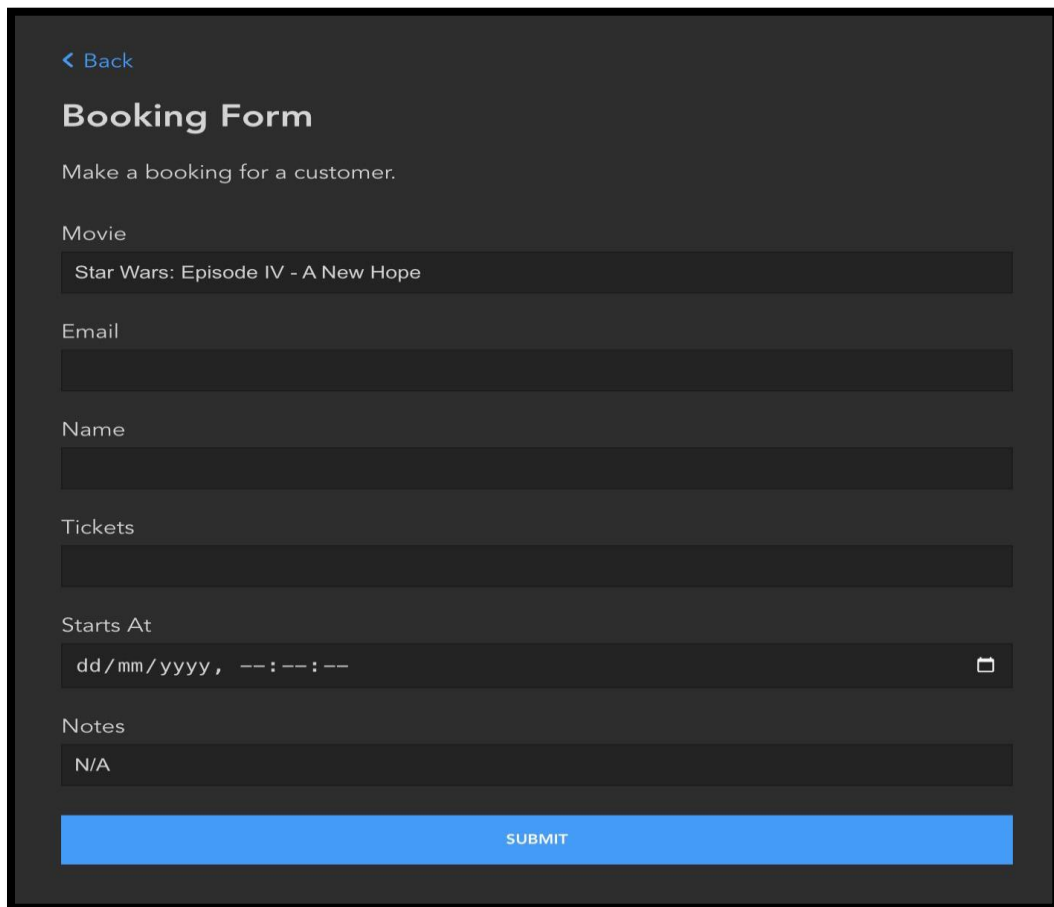
This module confirms the ticket after checking seat availability.

5. Database Module

This module stores confirmed booking details in the MySQL database.

6.6 System Screenshots / Output Screens

Home Page



The screenshot displays a 'Booking Form' on a dark-themed background. At the top left, there is a '< Back' link. The title 'Booking Form' is centered, followed by the instruction 'Make a booking for a customer.' The form contains several input fields: 'Movie' with the value 'Star Wars: Episode IV - A New Hope', 'Email', 'Name', 'Tickets', and 'Starts At' with a date/time placeholder 'dd/mm/yyyy, --:--:--' and a calendar icon. Below these is a 'Notes' field with the value 'N/A'. At the bottom, a prominent blue 'SUBMIT' button is centered.

Movie Selection Screen

Name *: _____ Number of Seats *: _____

SCREEN

	1	2	3	4	5	6	7	8	9	10	11	12
A	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
B	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
C	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
D	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
E	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
F	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
G	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
H	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
I	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
J	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

■ Selected Seat
 ■ Reserved Seat
 ■ Empty Seat

Name	Number of Seats	Seats

Seat Selection Screen

Select Seats

THU 17 DEC

10:00 11:30 2:30 **4:45** 7:20 11:10

● Not Available ● Available ✓ Selected ● Purchased

Row	1	2	3	4	5	6	7	8	9	10	11	12
A	●	●	●	●	●	●	●	●	●	●	●	●
B	●	●	●	●	●	●	●	●	●	●	●	●
C	●	●	●	●	●	●	●	●	●	●	●	●
D	●	●	●	●	●	●	●	●	●	●	●	●
E	●	●	●	●	●	●	●	●	●	●	●	●
F	●	●	●	●	●	●	●	●	●	●	●	●
F	●	●	●	●	●	●	●	●	●	●	●	●
G	●	●	●	●	●	●	●	●	●	●	●	●
H	●	●	●	●	●	●	●	●	●	●	●	●

Screen this way

3 Tickets
series 5, seats - F1, F2, F3

Total Amount : BD 6.60

PROCEED

Booking Confirmation Screen



6.7 Summary

The implementation of the **Online Movie Ticket Booking System** is completed using Python, Tkinter, and MySQL. All modules are integrated properly to provide a smooth and efficient ticket booking experience.

7. Testing

7.1 Introduction

Testing is an important phase in software development that ensures the system works correctly and meets all specified requirements. It helps in identifying errors, improving performance, and ensuring reliability of the system.

The **Online Movie Ticket Booking System** is tested to verify that all modules such as movie selection, seat booking, and database storage function properly without errors.

7.2 Testing Strategy

Different types of testing are performed to ensure proper functioning of the system. The main focus is on validating user input, checking seat availability, verifying booking confirmation, and ensuring proper database storage.

The following testing techniques are used:

- Unit Testing
 - Integration Testing
 - System Testing
-

7.3 Unit Testing

Unit testing is used to test individual components of the system separately.

Modules Tested:

- User Input Module
- Movie Selection Module
- Seat Selection Module
- Booking Module
- Database Module

Each module is tested independently to ensure correct functionality.

7.4 Integration Testing

Integration testing ensures that all modules work together properly after combining them.

Integration Checked:

- Frontend with Backend
- Backend with Database
- Seat Selection with Booking Module

This testing ensures smooth communication between all system components.

7.5 System Testing

System testing is performed to evaluate the complete working of the application.

Parameters Checked:

- Correct movie selection
- Proper seat booking
- Accurate booking confirmation
- Database storage of records
- System response time

The system performs all operations successfully without errors.

7.6 Test Case Table

Test Case ID	Input	Expected Output	Actual Output	Status
TC01	Valid user input	Booking successful	Booking done	Pass
TC02	Empty input	Error message	Error shown	Pass
TC03	Already booked seat	Seat not available	Warning shown	Pass
TC04	Valid booking	Data stored in DB	Stored successfully	Pass
TC05	Booking request	Confirmation screen	Displayed	Pass

7.7 Result Analysis

After performing all testing procedures, the system was found to work correctly and efficiently. The application successfully handles user input, processes booking requests, checks seat availability, and stores booking data in the database.

The system is reliable, user-friendly, and performs all required functions without any major errors.

7.8 Summary

Testing confirms that the **Online Movie Ticket Booking System** is functioning properly and meets all requirements. The system is ready for practical use and provides an efficient solution for ticket booking.

8. Result and Discussion

8.1 Introduction

This chapter presents the results obtained after the successful implementation and testing of the **Online Movie Ticket Booking System**. It evaluates how effectively the system performs ticket booking operations and discusses the overall performance, usability, and reliability of the system.

8.2 System Results

The developed system successfully performs all the required operations related to movie ticket booking. The application allows users to enter their details, select movies, choose show timings, and book seats efficiently.

The output generated by the system includes:

- Display of available movies and show timings
- Selection of seats with availability status
- Booking confirmation message
- Storage of booking details in database

The system provides quick and accurate results, ensuring a smooth user experience. All the modules are integrated properly and work together without any issues.

8.3 Performance Evaluation

The performance of the system is evaluated based on different parameters:

1. Speed and Efficiency

The system processes user input quickly and generates booking confirmation within a short time. There is no noticeable delay in seat selection or booking confirmation.

2. Accuracy

The system accurately checks seat availability and prevents double booking. All booking details are stored correctly in the database.

3. Usability

The interface is simple and easy to use. Users can navigate through the system without any technical knowledge.

4. Reliability

The system performs consistently without errors. It handles both valid and invalid inputs effectively.

5. Database Performance

The MySQL database efficiently stores and retrieves booking data without data loss or corruption.

8.4 Discussion

The **Online Movie Ticket Booking System** provides a practical solution to the problems associated with traditional ticket booking methods. It eliminates the need for standing in long queues and allows users to book tickets conveniently from their system.

The integration of Python (backend), Tkinter (frontend), and MySQL (database) ensures smooth functioning of the system. The modular design allows easy maintenance and future improvements.

One of the major advantages of the system is its simplicity. Unlike large commercial platforms, this system focuses on core functionalities, making it suitable for educational purposes and small-scale implementations.

However, the system has some limitations. It does not include online payment features and is limited to a basic interface. These limitations can be addressed in future versions by integrating advanced technologies.

8.5 Limitations

- No online payment integration
 - Limited to desktop-based application
 - Basic user interface
 - Requires manual database setup
-

8.6 Summary

The results show that the **Online Movie Ticket Booking System** is efficient, reliable, and user-friendly. It successfully meets the objectives of the project and provides a simple solution for movie ticket booking. The system performs well under different conditions and can be further enhanced in the future.

9. Features of the System

The **Online Movie Ticket Booking System** provides several important features that make it efficient, user-friendly, and reliable. The main features of the system are as follows:

9.1 User-Friendly Interface

The system provides a simple and interactive graphical user interface using Tkinter. Users can easily navigate through different options without requiring technical knowledge.

9.2 Movie Selection Facility

Users can view a list of available movies and select their preferred movie along with suitable show timings.

9.3 Seat Selection System

The system displays available seats in a structured format. Users can choose seats according to availability, and already booked seats are restricted.

9.4 Real-Time Seat Availability

The system checks seat availability in real time and prevents double booking, ensuring accurate results.

9.5 Booking Confirmation

After successful booking, the system displays a confirmation message with booking details such as movie name, seat number, and show timing.

9.6 Database Storage

All booking details are stored securely in a MySQL database, allowing future retrieval and record maintenance.

9.7 Input Validation

The system validates user inputs to ensure correct data entry and avoids errors during booking.

9.8 Fast Processing

The system processes user requests quickly and provides instant results, improving efficiency.

9.9 Error Handling

The system handles invalid inputs and unavailable seats by displaying appropriate error messages.

9.10 Scalable Design

The system is designed in a way that allows future enhancements such as online payment integration, mobile app support, and real-time updates.

9.11 Automated Booking System

The system automates the entire booking process, reducing manual effort and saving time for users.

10. Hardware and Software Requirements

10.1 Introduction

Hardware and software requirements define the resources needed for the development and smooth functioning of the **Online Movie Ticket Booking System**. The system is designed to run on standard computer systems without requiring high-end configurations.

10.2 Hardware Requirements

The project can be developed and executed on a basic computer system.

Minimum Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB
Hard Disk	20 GB free space
Display	1024 × 768 resolution
Input Devices	Keyboard, Mouse

Recommended Hardware Requirements

Component	Specification
Processor	Intel Core i5 / i7
RAM	8 GB or higher
Storage	SSD (for better performance)
Display	Full HD (1920 × 1080)

10.3 Software Requirements

The system is developed using programming and database tools.

Operating System

- Windows 10 / Windows 11
 - Linux
 - macOS
-

Frontend Technology

- **Tkinter** – Used for graphical user interface design
-

Backend Technology

- **Python 3.x** – Used for application logic and processing
-

Database

- **MySQL** – Used for storing booking records
-

Development Tools

- **Visual Studio Code (VS Code)** – Code editor
 - **Python IDLE / Anaconda** – Python execution environment
 - **XAMPP (optional)** – Database management
-

Libraries and Modules

- Tkinter – GUI development
 - mysql.connector – Database connectivity
-

10.4 Summary

The hardware and software requirements of the system are simple and easily available. The use of open-source technologies makes the system cost-effective and easy to implement.

11. Conclusion

The **Online Movie Ticket Booking System** has been successfully designed and implemented to provide an efficient and user-friendly solution for booking movie tickets. The system simplifies the traditional ticket booking process by allowing users to select movies, choose show timings, and book seats through a digital platform.

The application effectively integrates **Python (backend)**, **Tkinter (frontend)**, and **MySQL (database)** to ensure smooth functioning and accurate data handling. It successfully performs all the required operations such as displaying movies, checking seat availability, processing bookings, and storing booking details in the database.

The system reduces manual effort, saves time, and eliminates the need for standing in long queues at cinema halls. It also ensures accuracy by preventing double booking and providing real-time seat availability. The user-friendly interface makes the system easy to use, even for non-technical users.

Overall, the project demonstrates the practical application of software development concepts in solving real-world problems. It meets all the objectives defined at the beginning and provides a reliable and efficient ticket booking solution.

In conclusion, the **Online Movie Ticket Booking System** is a successful implementation of a simple yet effective application that enhances user convenience and improves the ticket booking experience.

12. Future Scope

The **Online Movie Ticket Booking System** developed in this project provides a basic and efficient solution for booking movie tickets. However, there are several opportunities to enhance and expand the system in the future to improve its functionality, usability, and performance.

In future, the system can be upgraded by integrating an **online payment gateway**, which will allow users to make secure payments using debit cards, credit cards, or digital wallets. This will make the booking process completely digital and more convenient.

The system can also be converted into a **web-based or mobile application**, enabling users to access the booking platform from anywhere at any time. This will increase accessibility and usability across different devices such as smartphones and tablets.

Another important improvement would be the addition of **real-time data integration with cinema servers**, which will provide accurate and updated information about available movies, show timings, and seat availability. This will make the system more reliable and efficient.

Advanced features such as **user login and registration, booking history, and personalized recommendations** can also be included to enhance user experience. The system can use Artificial Intelligence techniques to suggest movies based on user preferences and past bookings.

The seat selection interface can be further improved by adding **visual seat layouts with color coding**, making it easier for users to identify available and booked seats.

Additionally, the system can include features like **email or SMS ticket confirmation**, which will provide users with booking details instantly. Integration with QR code-based tickets can also be implemented for faster entry at cinema halls.

Overall, the system has great potential for future development and can be transformed into a complete commercial-level application by incorporating advanced technologies and features.

13. References

The following references were used during the development of the **Online Movie Ticket Booking System**. These sources provided valuable information related to programming concepts, database management, and system development:

1. **Python Documentation**, Python Software Foundation.
Available at: <https://www.python.org/doc/>
2. **MySQL Official Documentation**, Oracle Corporation.
Available at: <https://dev.mysql.com/doc/>
3. **Tkinter GUI Programming Guide**, Python Library Reference.
Available at: <https://docs.python.org/3/library/tkinter.html>
4. **W3Schools Tutorials** (HTML, CSS, JavaScript Basics).
Available at: <https://www.w3schools.com>
5. **GeeksforGeeks**, Programming and Database Concepts.
Available at: <https://www.geeksforgeeks.org>
6. **Stack Overflow**, Developer Community Discussions.
Available at: <https://stackoverflow.com>
7. **Sweigart, Al.**, *Automate the Boring Stuff with Python*, No Starch Press, 2015.
8. **Ramakrishnan, R., and Gehrke, J.**, *Database Management Systems*, McGraw-Hill Education.
9. **Silberschatz, A., Korth, H. F., and Sudarshan, S.**, *Database System Concepts*, McGraw-Hill.
10. **Pressman, Roger S.**, *Software Engineering: A Practitioner's Approach*, McGraw-Hill.

14. Appendix

Appendix A – Source Code

Python Code (Main Program)

```
import tkinter as tk
from tkinter import messagebox
import mysql.connector

# Database Connection
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="",
    database="movie_db"
)
cursor = conn.cursor()

def book_ticket():
    name = entry_name.get()
    movie = movie_var.get()
    time = time_var.get()
    seat = entry_seat.get()
```

```

# Check empty fields

if name == "" or seat == "":

    messagebox.showerror("Error", "All fields are required")

    return


# Check if seat already booked

cursor.execute("SELECT * FROM booking WHERE seat_no=%s", (seat,))

result = cursor.fetchone()


if result:

    messagebox.showwarning("Warning", "Seat already booked")

else:

    cursor.execute("INSERT INTO booking (name, movie, show_time, seat_no) VALUES
(%s,%s,%s,%s)",

                    (name, movie, time, seat))

    conn.commit()

    messagebox.showinfo("Success", "Ticket Booked Successfully")


# GUI

root = tk.Tk()

root.title("Movie Booking System")


tk.Label(root, text="Name").pack()

entry_name = tk.Entry(root)

entry_name.pack()

```

```

tk.Label(root, text="Select Movie").pack()

movie_var = tk.StringVar()

tk.OptionMenu(root, movie_var, "Movie 1", "Movie 2", "Movie 3").pack()


tk.Label(root, text="Select Time").pack()

time_var = tk.StringVar()

tk.OptionMenu(root, time_var, "10 AM", "2 PM", "6 PM").pack()


tk.Label(root, text="Seat No").pack()

entry_seat = tk.Entry(root)

entry_seat.pack()


tk.Button(root, text="Book Ticket", command=book_ticket).pack()


root.mainloop()

```

Database Connection Code (MySQL)

```

import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="",
    database="movie_db"
)

cursor = conn.cursor()

```

Appendix B – Database Details

[🔗 Database Creation](#)

```
CREATE DATABASE movie_db;  
USE movie_db;
```

[🔗 Table Creation](#)

```
CREATE TABLE booking (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(50),  
    movie VARCHAR(100),  
    show_time VARCHAR(50),  
    seat_no VARCHAR(20)  
);
```